



University
of Exeter

Lectures 13 & 14 – Multi-objective (Evolutionary) Algorithms

ECM3412/ECMM409 – Nature-Inspired Computation

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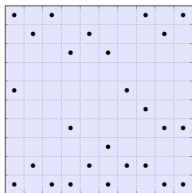
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Example: offshore wind farm layouts



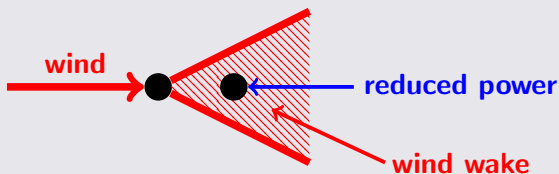
- Want to simultaneously **minimise cost** and **maximise power output**
- Hard to do – there is a **trade-off** between the two objectives
 - Maximising energy production is expensive
 - The cheapest solution minimises energy production
- Use a **multi-objective GA** to optimise both objectives together and **return a set of solutions**



P. L. Manikowski, D. J. Walker & M. J. Craven. "Multi-Objective Optimisation of the Benchmark Wind Farm Layout Problem", J. Mar. Sci. Eng., 9(12):1376, 2021.

Wind wake effect

A turbine in the shadow (**wind wake**) of another turbine generates less power



Multi-objective optimisation problem definition

Given a solution $\mathbf{x}$, whereas single-objective quality is $y = f(\mathbf{x})$ a multi-objective solution's quality is determined by an **objective vector**:

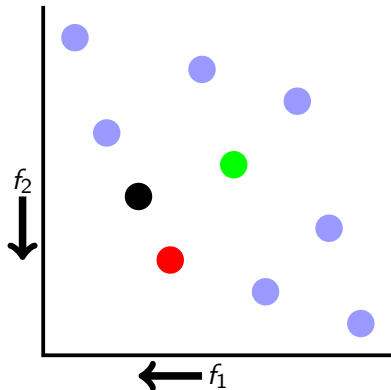
$$\mathbf{y} = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_M(\mathbf{x})),$$

subject to any feasibility constraints and where M is the number of objectives

How can we get EAs to perform multi-objective optimisation?

- Generational GA
- Revised objective functions: **multiple objectives**
- Revised selection criteria: **dominance** criterion
- Revised visualisation: Pareto front approximations (**curves/surfaces**)

A solution **a** **dominates** another **b** in the population if it is at least as good as **b** on every objective and better than **b** on at least one objective

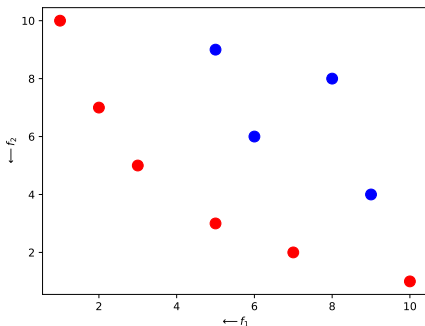


Assuming both objectives are to be minimised

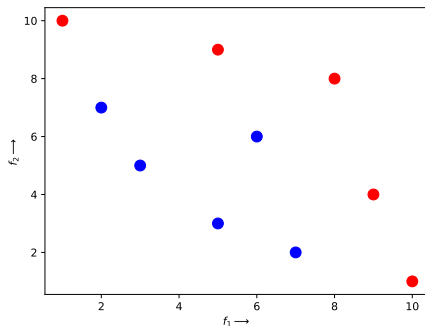
- **Red** dominates **green** because it has smaller (better) values in both objectives
- **Black** also dominates **green**
- Neither **red** or **black** dominates the other – **black** is better on f_1 and **red** is better on f_2
- **Red** and **black** are called **mutually non-dominating**

Given the dominance relation we know that the best solutions will lie along a curve consisting of non-dominated points – this is known as the Pareto front

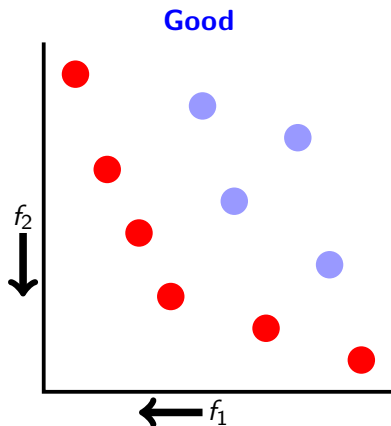
Minimise f_1 and f_2



Maximise f_1 and f_2

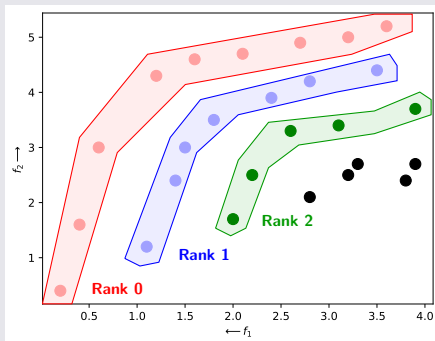


- Evenly-spaced solutions
- Covering the largest possible area of the front



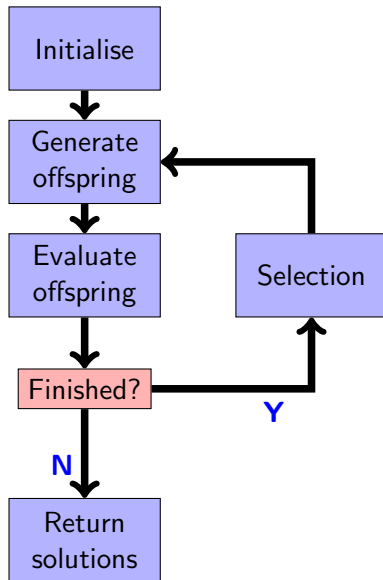
- Instead of having a single objective value we have **multiple fitnesses**
- We know that the best solutions lie on the Pareto front
- How do we discriminate between the rest?
- We can use **ranking methods**

Non-dominated sorting



- Find the non-dominated solutions (**Rank 0**) and remove them from the population
- Find the next non-dominated set (**Rank 1**), discard those
- Find the next non-dominated set (**Rank 2**), discard those
- and so on...

- Crossover and mutation can remain broadly the same
- With ranking we can tell which solutions are best (non-dominated) and which are worst
- How can we tell the difference between two solutions which are non-dominated?
- We need a **new selection operator**



Basic tournament selection with Pareto dominance:

Select two random individuals from the population, if one dominates the other then select it

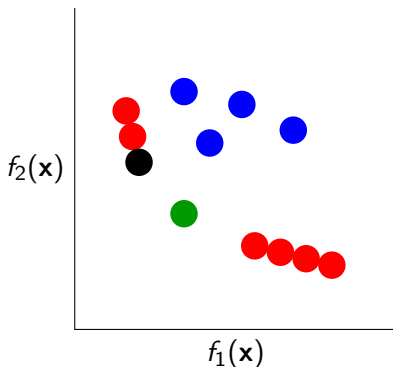
Selection pressure is not sufficient – solutions chosen may be mutually non-dominating in which case random choice is made

Pareto dominance tournament selection

- 1 Select two random solutions $\mathbf{x}$ and $\mathbf{y}$
- 2 Also select a separate comparison set $\mathbf{Z} = \{\mathbf{z}_i\}_{i=1}^C$
- 3 If $\mathbf{x}$ or $\mathbf{y}$ is non-dominated with respect to all $\mathbf{z}_i \in \mathbf{Z}$ then select
- 4 If $\mathbf{x}$ and $\mathbf{y}$ have the same domination then select at random (tiebreak)

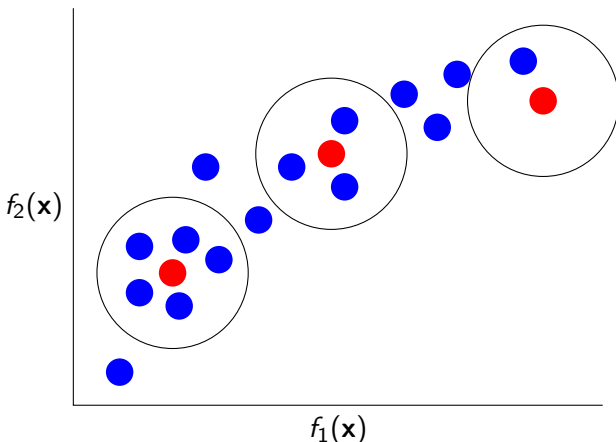
The value of C is used to control selection pressure

- A problem with Pareto dominance-based tournament selection is that two individuals can be mutually non-dominated
- We need a way of breaking ties between such individuals that ensures a good spread of solutions



- If the black and green solutions are selected by the tournament then both are non-dominated – regardless of the comparison set
- The green solution should be selected as it occupies a less densely-populated area of the space

- Separate the fitness landscape or genotype into **niches**
- Prefer individuals in a niche with fewer individuals
- New parameter – **niche radius**



- Elitist MOGA
- Generates new solution using selection, crossover and mutation
- Uses a **fast non-dominated sort** to rank the solutions
- Uses **crowding distance** as a tie breaker
- Cited tens of thousands of times

K. Deb, A. Pratap, S. Agarwal and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Transactions on Evolutionary Computation*, 6(2):182–197, 2002.

A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II

Kalyanmoy Deb, Associate Member, IEEE, Amrit Pratap, Sameer Agarwal, and T. Meyarivan

Abstract—Multiobjective evolutionary algorithms (EAs) that use nondominated sorting and sharing have been criticized mainly for their: 1) $O(MN^2)$ computational complexity (where M is the number of objectives and N is the population size); 2) nondomination approach; and 3) the need for specifying a sharing parameter. In this paper, we suggest a nondominated sorting-based multiobjective EA (MOEA), called nondominated sorting genetic algorithm II (NSGA-II), which alleviates all the above three difficulties. Specifically, a fast nondominated sorting approach with $O(MN)$ computational complexity is presented. Also, a selection operator is presented that creates a mating pool by combining the parent and offspring populations and selecting the best (with respect to fitness and spread) N solutions. Simulation results on difficult test problems show that the proposed NSGA-II, in most problems, is able to find much better spread of solutions and better convergence near the true Pareto-optimal front compared to Pareto-archived evolution strategy and strength-Pareto EA—two other elitist MOEAs that pay special attention to creating a diverse Pareto-optimal front. Moreover, we modify the definition of dominance in order to save constrained multiobjective problems efficiently. Simulation results of the constrained NSGA-II on a number of test problems including a five-objective seven-constraint nonlinear problem, are compared with another constrained multiobjective optimizer and much better performance of NSGA-II is observed.

Index Terms—Constraint handling, elitism, genetic algorithms, multicriterion decision making, multiobjective optimization, Pareto-optimal solutions.

I. INTRODUCTION

THE PRESENCE of multiple objectives in a problem, in principle, gives rise to a set of optimal solutions (largely known as Pareto-optimal solutions), instead of a single optimal solution. In the absence of any further information, one of these Pareto-optimal solutions cannot be said to be better than the other. This demands a user to find as many Pareto-optimal solutions as possible. Classical optimization methods (including the multicriterion decision-making methods) suggest converting the multiobjective optimization problem to a single-objective optimization problem by emphasizing one particular Pareto-optimal solution at a time. When such a method is to be used for finding multiple solutions, it has to be applied many times, hopefully finding a different solution at each simulation run.

Over the past decade, a number of multiobjective evolutionary algorithms (MOEAs) have been suggested [1], [7], [13].

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Publisher Item Identifier 10.1109/78X028102101-2.

[20], [26]. The primary reason for this is their ability to find multiple Pareto-optimal solutions in one single simulation. Since evolutionary algorithms (EAs) work with a population of solutions, a simple EA can be extended to maintain a diverse set of solutions. With an emphasis for moving toward the true Pareto-optimal region, an EA can be used to find multiple Pareto-optimal solutions in one single simulation run.

The nondominated sorting genetic algorithm (NSGA) proposed in [20] was one of the first such EAs. Over the years, the main criticisms of the NSGA approach have been as follows.

- 1) *High computational complexity of nondominated sorting*: The currently-used nondominated sorting algorithm has a computational complexity of $O(MN^2)$ (where M is the number of objectives and N is the population size). This makes NSGA computationally expensive for large population sizes. This large complexity arises because of the complexity involved in the nondominated sorting procedure in every generation.
- 2) *Lack of elitism*: Recent results [25], [18] show that elitism can speed up the performance of the GA significantly, which also can help preventing the loss of good solutions once they are found.
- 3) *Need for specifying the sharing parameter* σ_{share} : Traditional mechanisms of ensuring diversity in a population so as to get a wide variety of equivalent solutions have relied mostly on the concept of sharing. The main problem with sharing is that it requires the specification of a sharing parameter (σ_{share}). Though there has been some work on dynamic sizing of the sharing parameter [10], a parameter-less diversity-preservation mechanism is desirable.

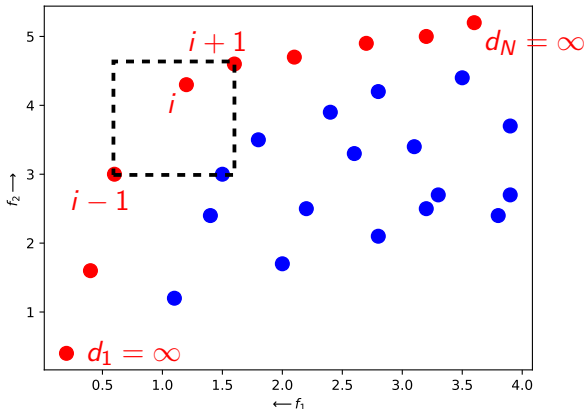
In this paper, we address all of these issues and propose an improved version of NSGA, which we call NSGA-II. From the simulation results on a number of difficult test problems, we find that NSGA-II outperforms two other contemporary MOEAs: Pareto-archived evolution strategy (PAES) [14] and strength-Pareto EA (SPEA) [24] in terms of finding a diverse set of solutions and in converging near the true Pareto-optimal set.

Constrained multiobjective optimization is important from the point of view of practical problem solving, but not much attention has been paid so far in this respect among the EA researchers. In this paper, we suggest a simple constraint-handling strategy with NSGA-II that suits well for any EA. Our test problems chosen from the literature, NSGA-II has been compared with another recently suggested constraint-handling strategy. These results encourage the application of NSGA-II to more complex and real-world multiobjective optimization problems.

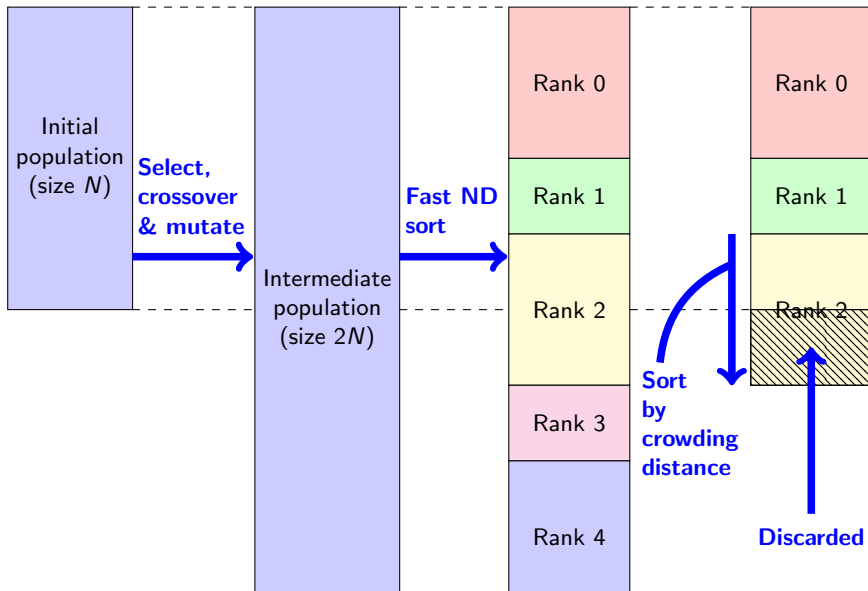
In the remainder of the paper, we briefly mention a number of existing elitist MOEAs in Section II. Thereafter, in Section III,

crowding_distance(Y)

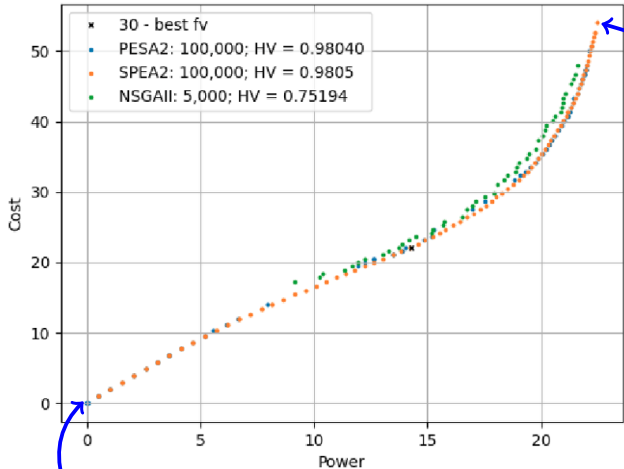
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1:  $N = |\mathbf{Y}|$ 
2: for  $i = 1, \dots, N$  do
3:    $d_i = 0$ 
4: end for
5: for  $m = 1, \dots, M$  do
6:    $\mathbf{Y} = \text{sort}(\mathbf{Y}, m)$ 
7:    $d_1 = d_N = \infty$ 
8:   for  $i = 2, \dots, (N - 1)$  do
9:      $d_i = d_i + (\mathbf{Y}_{i+1}^m - \mathbf{Y}_{i-1}^m) / (f_m^{\max} - f_m^{\min})$ 
10:  end for
11: end for
```



NSGA-II execution



NSGA-II example: offshore wind

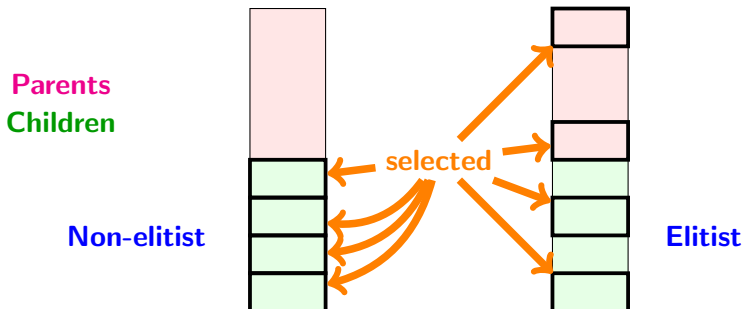


Best power output, most expensive

Best possible solution – but infeasible

Cheapest, worst power output

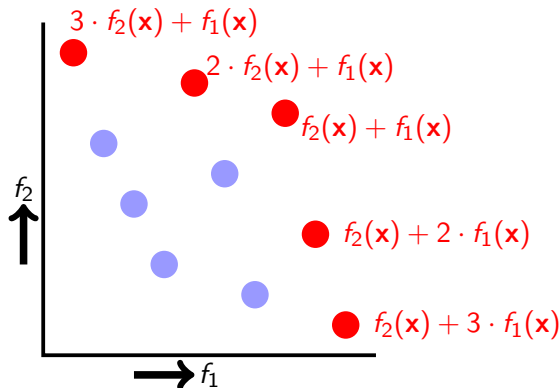
P. L. Manikowski, D. J. Walker, M. J. Craven, Multi-Objective Optimisation of the Benchmark Wind Farm Layout Problem. *Journal of Marine Science and Engineering* 9(12):1376, 2021



Elitist MOGAs (PAES, SPEA, NSGA-II) have somewhat superseded non-elitist versions – **Why?**

- NSGA-II requires no extra parameters (e.g. niche sizes)
- Executes and converges faster
- Some evidence that on certain problems it can prematurely converge

- The population-based approach of GAs is ideal for multi-objective optimisation
- Could we use single-objective EAs? **Yes** by weighting each objective we can optimise for different points



But...

- Have to run a separate GA to find each point
- Regions of some Pareto front geometries will be missed

Three issues with many-objective optimisation

- 1 Problems with **visualising** a high-dimensional Pareto front approximation
- 2 **Reduction in the search** capability of multi-objective optimisers
- 3 Exponential **increase in the number of solutions needed** to approximate the Pareto front

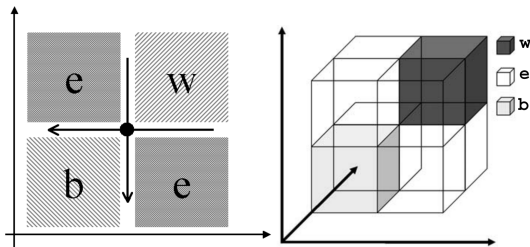
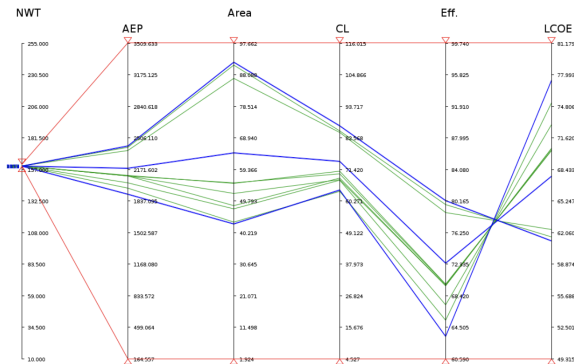


Fig. 1. Schematic view of Pareto-dominance based partial order in 2D and 3D problems when a candidate solution is considered (•): equal (e), better (b) and worse (w) solution regions are shown.

H. Ishibuchi, N. Tsukamoto & Y. Nojima, "Evolutionary Many-objective Optimisation: A Short Review", Proc. IEEE Congress on Evolutionary Computation, 2008

M. Farina & P. Amato, "Fuzzy Optimality and Evolutionary Multiobjective Optimization", Proc. Int. Conference on Evolutionary Multi-criterion Optimisation, 2003

- Six objectives: **minimise** (1) # turbines; (2) area; (3) annual energy prediction; (4) levelised cost of energy; **maximise** (5) annual energy production; (6) efficiency
- Optimise with NSGA-III (many-objective optimiser)
- Visualise with **parallel coordinate plot**



Multi-objective optimisation

- Optimise a problem comprising **two or more** conflicting objectives
- Compare solutions using **Pareto dominance**
- Goal is to find the **Pareto front**

Multi-objective EAs

- Replace solution comparison with dominance
- Drive population towards Pareto front and **maintain diversity**

Many-objective optimisation

- ≥ 4 objectives
- Cannot simply rely on dominance – solutions are incomparable
- Visualisation challenge